Combining the two terms in (6.13) and (6.14) we have that

$$\Delta\omega = \frac{\omega_0}{c^2}\left(gH - \frac{v^2}{2}\right). \tag{6.15}$$

Such a frequency shift of our moving clock means that if we measure a time dt on a fixed clock, the moving clock will register the time

$$dt\left[1 + \left(\frac{gH}{c^2} - \frac{v^2}{2c^2}\right)\right]. \tag{6.16}$$

The total time excess over the trajectory is the integral of the extra term with respect to time, namely

$$\frac{1}{c^2}\int\left(gH - \frac{v^2}{2}\right)dt, \tag{6.17}$$

which is supposed to be a maximum.

The term gH is just the gravitational potential ϕ. Suppose we multiply the whole thing by a constant factor $-mc^2$, where m is the mass of the object. The constant won't change the condition for the maximum, but the minus sign will just change the maximum to a minimum. Equation (6.16) then says that the object will move so that

$$\int\left(\frac{mv^2}{2} - m\phi\right)dt = \text{ a minimum.} \tag{6.18}$$

But now the integrand is just the difference of the kinetic and potential energies. And if you look in Chapter 19 of Volume II* you will see that when we discussed the principle of least action we showed that Newton's laws for an object in any potential could be written exactly in the form of Eq. (6.18).

6-9 Einstein's theory of gravitation

Einstein's form of the equations of motion—that the proper time should be a maximum in curved space-time—gives the same results

* of *The Feynman Lectures on Physics*.